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PROFESSIONAL SUMMARY

Software Engineer and ML practitioner with 5+ years of Python experience, 2+ years of C++ development, and an M.Sc. in Data Science & Artificial Intelligence. Proven track record in voice AI (NLU domain engineering, speech recognition), LLM integration, backend API development, and CI/CD automation. Experienced in delivering production-grade systems at AI companies in Germany.

TECHNICAL SKILLS

Languages (Programming)	Python, C++, Bash, JavaScript, TypeScript, Java, R, HTML
ML / AI	PyTorch, scikit-learn, XGBoost, NLTK, spaCy, Transformers, OpenAI API, Anthropic API, Mistral, LLaMA
NLP / Speech	NLU domain engineering, speech recognition pipelines, Tamil NLP, audio data processing, LLM-based code generation (CodeWeaver)
Backend / APIs	FastAPI, Django, RESTful API design, MongoDB, SQL, Prisma
Frontend / UI	React, Next.js, Vue, Node.js, Tailwind CSS
DevOps / CI-CD	Jenkins, GitLab CI, Docker (familiar), Git, GitLab, GitHub
Data	Pandas, NumPy, feature engineering, regression, XGBoost
Cloud / Tools	Azure (familiar), n8n, Linux CLI (advanced)
Human Languages	English (professional fluency), German (B1), Tamil (native)

WORK EXPERIENCE

SoundHound AI, Berlin, Germany Working Student Software Engineer

10/2023 – 09/2025

Voice AI / C++ / CodeWeaver / Gitlab / CI/CD

- Engineered multiple production NLU voice-AI domains in Tamil using C++, enabling natural-language understanding for millions of regional users
- Leveraged **LLM-based code generation** (CodeWeaver / GPT, Mistral, LLaMA) to automatically produce car-control grammar domains, cutting manual authoring time significantly
- Collected, cleaned, and processed large-scale Tamil speech datasets from multiple sources to train and validate speech recognition models
- Built an automatic **declension generator** for Tamil nouns and verbs, improving linguistic coverage of NLU grammars
- Conducted **comparative LLM benchmarking** (GPT-4, Mistral, LLaMA) across Indian languages to select optimal models for production use
- Delivered internal seminars on CodeWeaver; conducted **technical interviews** for student recruits and onboarding sessions for new team members
- Collaborated with cross-language teams to standardise localisation workflows and improve consistency across 4 Indian language projects

Natif ai, Saarbrücken, Germany Working Student Backend Engineer

07/2021 – 09/2023

Python / ML / Backend Fast API / OCR / CI/CD / Gitlab

- Extended a proprietary **Python testing framework** for deep-learning document extraction pipelines, improving

pipeline evaluation performance by **35 %**

- Designed and implemented a document-extraction aggregation framework supporting diverse product and customer requirements
- Built scalable **FastAPI + MongoDB** backend services powering document intelligence pipelines in production
- Developed custom extraction use cases in collaboration with domain experts, annotation teams, and customer-success stakeholders
- Participated in database-schema design seminars to generalise schema across heterogeneous document types and use cases

Wipro Ltd, Bengaluru, India

Software Engineer

07/2019 – 09/2020

Python / Bash / DevOps / Server Security

- Automated security hardening tasks for server migration to **RHEL 7**, reducing manual compliance effort and ensuring audit readiness
- Built and maintained a **CI/CD pipeline with Jenkins** — integrating automated tests, deployment scripts, and monitoring for the Ericsson project
- Documented automation workflows and conducted knowledge-transfer sessions to improve team-wide process efficiency

EDUCATION

M.Sc. Data Science & Artificial Intelligence

2021 – 2026

Saarland University, Saarbrücken, Germany

GPA: 2.4

B.E. Computer Science & Engineering

2015 – 2019

Anna University, Chennai, India

GPA: 8.1 / 10.0

PERSONAL PROJECTS

CouplesOnly - Couples Planning & Tracking Platform

[\[View GitHub\]](#)

Tech Stack: *Next.js 14, PostgreSQL, Tailwind CSS, Python*

- Full-stack lifestyle platform for couples to plan and track day-to-day life together
- Features: partner linking with approval system, **Wedding Planner**, anniversary tracking with countdown and Polaroid-style photo uploads, shared expense & health tracking, todo lists, and a photo gallery
- Architecture: Next.js API Routes (serverless) for backend, Python micro-workers for link extraction, PostgreSQL as primary database
- Wedding Planner and Anniversary Space modules currently in **beta**

Retro Developer Portfolio

[\[View GitHub\]](#)

Tech Stack: *Django, Python, JavaScript, CSS, PostgreSQL*

- Personal portfolio with a fully custom **CRT / 90s retro aesthetic**: pixel fonts, scan-line effects via **repeating-linear-gradient**, and dark colour scheme
- **Database-driven content** (no static rendering) for easy admin management
- Hand-crafted **multilingual support** (English, German, Tamil) stored in session without machine translation; language switchable via URL parameter (**?lang=en**)
- Category-based navigation (C++ Developer, Python Developer, ML Engineer, CI/CD, Skills) with a live **command-line terminal** that auto-updates displayed skills on page navigation via JavaScript

Tamil NMT & ASR from Scratch

[\[View GitHub\]](#)

Tech Stack: *Python, PyTorch, Transformer, Deep Speech 2*

- Built a **Neural Machine Translation** model for Tamil–German translation from scratch using the Transformer architecture in PyTorch, without pre-trained weights
- Implemented an **ASR system** for Tamil using the **Deep Speech 2** architecture end-to-end in PyTorch
- Deepened expertise in CTC loss, low-resource language modelling, and sequence-to-sequence learning

DoorDash ETA Prediction - ML Regression

[\[View GitHub\]](#)

Tech Stack: *Python, XGBoost, Random Forest, Pandas*

- Built an end-to-end ML pipeline to predict delivery duration from historical order data
- Engineered temporal, geospatial, and contextual features for model input

- Evaluated Linear Regression, Random Forest, and XGBoost using RMSE, MAE, and cross-validation; selected best-performing model to reduce prediction error vs. baseline

ACADEMIC PROJECTS

M.Sc. Thesis - Tamil Discourse Parsing (PDTB 3.0)

[\[View GitHub\]](#)

Tech Stack: *Python, PyTorch, NLP, spaCy, Transformers, Fine Tuning, Adapters, Prompt Engineering*

- Created a novel **Tamil discourse dataset** from scratch — the first PDTB-3.0-annotated resource for Tamil, covering implicit and explicit discourse relations
- Trained and evaluated fine-tuned Transformer, LSTM-based classifier, and SVM baseline for automatic discourse parsing, benchmarked against PDTB 3.0 standard metrics
- Addressed low-resource NLP challenges: data scarcity, Tamil morphological complexity, and cross-lingual transfer from English-trained discourse models

Speaker-Independent Speech Digit Recognition (SDR)

[\[View GitHub\]](#)

Tech Stack: *PyTorch, Deep CNN, RNN-LSTM, Transformer*

- Developed a speaker-independent SDR system comparing **Deep CNN**, **RNN-LSTM**, and **Transformer** architectures on an audio-digit dataset
- Implemented entirely in **PyTorch**; gained hands-on expertise in **contrastive loss** and **self-supervised learning**

Reinforcement Learning - Bird-Hunt Game Agent

[\[View GitHub\]](#)

Tech Stack: *Python, Unity ml-agents, PPO, GAIL*

- Trained an autonomous shooting agent in a Unity 2D environment using **PPO**, **GAIL**, and **Behaviour Cloning** via Unity ml-agents

UniPocket - Student Web Application

[\[View GitHub\]](#)

Tech Stack: *ReactJS, Prisma, REST API, Node.js*

- Full-stack web app for students to discover available lectures and connect with peers
- Designed and implemented REST backend, Prisma ORM schema, and React frontend

CERTIFICATIONS

- R Programming for Data Science — Coursera
- Data Visualization with R — IBM